Muhammad Muizz Zafar

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Full-stack developer: C# / .NET Core REST APIs, Angular and React, SQL Server, and automated testing with Cypress and Jasmine.

Technical Skills

Programming Languages: C#, TypeScript, JavaScript, SQL, Python, Java, C++, Go
Tools & Frameworks: Angular, Angular Material, .NET Core, React, Vue.js, SQL Server, Cypress, Jasmine, Git

Education

University of Ottawa — Ottawa, ON
Bachelor of Applied Science, Software Engineering *GPA: 3.7/4.0* — Sept 2020 - Dec 2024

Work Experience

Software Analyst — May 2024 - Present
Elections Canada — Ottawa, ON

- Developed **C#/.NET Core** REST APIs backed by SQL Server databases, enabling internal teams to securely access and consume operational data through reusable endpoints.
- Developed a reusable ETL validation framework using **SQL**, **Python** and **Pandas**, enabling automated validation of large datasets (20M+ records) and created visual dashboards for data quality KPIs using **Streamlit**
- Built and maintained scalable ETL pipelines using Python, **PySpark**, Pandas, and SQL to extract data from Oracle databases, transform it, and load it into SQL Server data warehouse.
- Managed database ETL releases and coordinated version-controlled deployment scripts across DEV, UAT, and PROD, ensuring rollback capability while building **Power BI dashboards** to monitor KPIs, data quality metrics, and deployment status.

Full Stack Developer — September 2023 - August 2024
University of Ottawa — Ottawa, ON

- Optimized **SQL queries** and **LINQ-based C#/.NET REST API endpoints**, reducing event data lookup time by up to **15 seconds**.
- Revamped the event registration and admin pages by using **Angular** material UI components to improve UI layout and reducing redundant lines of code
- Redesigned the University of Ottawa's emergency alert website with **Angular** and **Typescript**, introducing a modern, interactive UI. Used by over **20,000** students daily.
- Wrote 9 **End to End tests** using **Cypress**, and **25 unit tests** using **Jasmine** for new APIs and features implemented on frontend and **NUnit** backend
- Collaborated in an **Agile sprint cycle** to review test results, ensure feature reliability, and fix UI/API bugs.

Cloud Developer Intern — Sep 2022 - Dec 2022 & May 2023 - Aug 2023
Trend Micro Inc. — Ottawa, ON

- Developed serverless data processing pipelines using AWS Lambda to analyze client files, images, and code stored in **Amazon S3**, performing vulnerability scans using Anchore APIs and Grype.
- Configured AWS CloudWatch and Splunk monitoring to track pipeline performance, logs, and failures, enhancing observability and system reliability.

Full Stack Developer Intern — January 2022 - April 2022
Applied Mind Inc. — Ottawa, ON

- Automated event detection workflows by integrating **Python**, **PostgreSQL**, and **GraphQL**, reducing manual QA verification time.
- Designed real-time dashboards in **Vue.js** for better monitoring and quicker validation of production events.
- Added dynamic updates to the graph, map, and notification components using **VUE** dynamic update feature and **GraphQL** subscriptions which improved UX

Full Stack Developer — June 2019 - June 2021
Tech Under Twenty — Oakville, ON

- Created and tested RESTful APIs using **.NET Core (C#)** for the member management system, improving stability and data validation accuracy.
- Designed responsive UI components in **Angular**, performing manual testing and debugging to improve user experience.

Projects

Hotel Room Booking System *React, Javascript, HTML, CSS* — **https://hotel-booking.github.io**

- Utilized React, Bootstrap, and Material UI to style the website frontend for the hotel booking website

Cycle Repair Booking System *Vue, Javascript, HTML, CSS* — **https://bikerepairs.github.io**

- Utilized Vue, to create the website for a cycle repair system using bootstrap grid system for a responsive design